



Financial Institutions & Markets- ENT 309

PPT #7, IITD, Summer, 2026

Agenda

1

FOREIGN EXCHANGE MARKET

- Types of FX transactions
- Which Interest Rate?
- Foreign Exchange Flows
- FX Demand & Supply, Equilibrium Exch Rate & Change in FX rates

Types of FX Transactions

FX Symbols



- € Euro
- JY¥/ 円 JPY/ in Kanji character
- CN¥ /元/RMB Chinese Yuan/ Renminbi
- ₪ Russian Ruble
- R\$ BRL Brazilian Real
- \$Ar Argentina Peso



Bid/ Offer or Ask

- What does USD/INR 94.0/94.05 mean?
 - At 94.0 Bank will buy USD (Bid) and sell INR and at 94.05 it will sell USD (Ask or Offer) and buy INR.
- USD/JPY 121.50/55. A customer wishing to buy JPY to the value of USD 1 million will receive JPY = ?
 - JPY 121,500,000.

Calculating Cross Rates

CROSS RATES: What is GBP/JPY? If USD/JPY & GBP/USD is given.

- USD/JPY 120.25/120.30 GBP/USD 1.5700/1.5705
- Bank buys USD & sells JPY at 120.25 and buys JPY at 120.30, and
- Bank buys GBP & sells USD at 1.5700 and sells GBP at 1.5705.

To determine the GBP/JPY cross rate

- **BID RATE:** Bank buys GBP and sells JPY
- $1.5700 \text{ (Buys 1 GBP \& Sells USD)} \times 120.25 \text{ (Buys 1 USD \& Sells JPY)} = 188.79 \text{ JPY/GBP (GBP/USD X USD/JPY)}$
- **OFFER RATE:** Bank sells GBP and buys JPY
- $1.5705 \text{ (Sells 1 GBP \& Buys USD)} \times 120.30 \text{ (Sell 1 USD \& Buys JPY)} = 188.93 \text{ JPY/GBP}$
- Hence GBP/JPY 188.79/188.93
- **Used for Arbitrage transactions-** arbitrage is the practice of taking advantage of a price difference between two or more markets

FX Market Transactions



Spot Transaction

- Two-day delivery transaction (T+2)
- Direct exchange between 2 currencies. Involves cash rather than a contract. No interest paid for 2 days.
- Currency exchange done at Spot Exchange Rate
- An American bank signed a contract on Monday for a transfer of €1,000,000 to a French bank in Paris. The spot rate was 1.2563 \$/ € .
- On Wednesday the US bank would transfer € 1,000,000 to Paris and the French bank would transfer \$1,256,300 to the US bank in New York at the same time.

FX Market Transactions



DERIVATIVE

Forward Transaction

- Currencies exchanged on a specific date in the future
- Buyer & Seller agree on a Future Transaction rate. A contract is done
- Trxn done at the agreed rate irrespective of what actual FX rate on that day is
- Forward could be for one day, few days, months or years
- Buyer assumes a "LONG" position. Buys today with a receipt in future.
- Seller goes "SHORT". Sells today with a delivery in future.

Example of How a Forward Contract Works

- ABC Factory in Scotland is looking to buy motorbikes from Taiwan. ABC agrees to pay USD \$500,000 to Taiwanese supplier in 3 months from now.

FX Market Transactions



DERIVATIVE

Forward Transaction

- 3 months Forward rate at the time of the deal is GBP £1.00 = USD \$1.32.
- ABC Factory therefore expects to pay GBP £378,788 for the equipment.
- In 3 months' time, Spot GBP £1.00 = USD \$1.25.
- Scenario 1: If ABC Factory doesn't use a Forward contract
 - In 3 months' time, when the business is ready to pay for the goods from Taiwan, they have to buy at Spot rate of GBP £1.00 = USD \$1.25. The goods now will cost £400,000.
 - ABC Factory would pay £21,212 more than anticipated originally.
- Scenario 2: ABC Factory uses a Forward contract
 - After 3 months, GBP £1.00 = USD \$1.25, But ABC Factory has a forward contract at GBP £1.00 = USD \$1.32. So they pay GBP £378,788 only.
 - Result ABC hedges & saves £21,212 due to the forward currency contract.

FX Market Transactions



DERIVATIVE

Forward Rate- INTEREST PARITY CONDITION

- Interest Parity Condition:
 - Assuming money moves across borders without any problems- there is capital mobility
 - If one currency pays higher interest rate, then everyone will invest in that currency only and zero in the other currency.
 - But people invest in both currencies - Both currencies are equally desirable by Investors
 - To have same Expected Returns in each currency, the interest differential will be reflected in the forward exchange rate.
 - Thus, the currency with a higher interest rate depreciates.

FX Market Transactions



DERIVATIVE

Forward Rate- INTEREST PARITY CONDITION

- A forward rate is not a forecast of what the exchange rate will be on the future date.
- It is the spot price adjusted for interest differential between the two currencies for the time period.
- Information required to calculate Forward Rate:
 - The interest rate of both currencies.
 - Number of days in the period.
 - Spot rate. Bid/Offer
 - Days per year: 360 (US, Japan, Europe etc) or 365 (UK, Aus, NZ)
- Why is it a derivative?
 - Value derived from Spot Rate & Interest Differential Between Two currencies

FX Market Transactions



DERIVATIVE

Forward Rate

- USD/JPY Spot: 120.25/120.50
- USD interest rates: 2.0% JPY interest rates: 0.5%
- Number of days: 30 Number of days in year for both currency: 360

BID FORWARD RATE

JPY 100 million at 120.25

30 days at 0.5%

Interest = 41667.00

Principal + Interest = 100,041,667

Equals

USD 831,600.83

30 days at 2.0%

Interest = 1386.00

Principal + Interest = 832,986.83

$\text{JPY} \div \text{USD} = 100,041,667 \div 832,986.83 = 120.10$

- This represents the 30 day forward Bid rate. USD/JPY = 120.10

FX Market Transactions



DERIVATIVE

Forward Rate

- USD/JPY Spot: 120.25/120.50 USD interest rates: 2.0%/JPY interest rates: 0.5%
- Number of days: 30 Number of days in year for both currency: 360

OFFER FORWARD RATE

JPY 100 million at 120.50

30 days at 0.5%

Interest = 41667.00

Principal + Interest = 100,041,667

EQUALS

USD 8,29,875.52

30 days at 2.0%

Interest = 1383.12

Principal + Interest = 831,258.64

$\text{JPY} \div \text{USD} = 100,041,667 \div 831,258.64 = 120.3496$

- This represents the 30 day forward OFFER rate. USD/JPY = 120.3496 (Approx. 120.35)

FX Market Transactions



DERIVATIVE

- USD/JPY Spot: 120.25/120.50 USD/JPY Forward: 120.10/ 120.35 (1 month)
JAP EXPORTER WANTS TO HEDGE USD PAYMENT HE WILL RECEIVE AFTER 1 MONTH
- He approaches a bank and says I will sell you USD after 1 month and I want to book it now
- Bank will buy at 1 month Forward Bid Rate: 120.10 JPY
- Bank will sell USD at offer rate today: 120.50 JPY
- Hedging cost: $120.50 - 120.10 = 0.40$ JPY. Exporter pays this cost and sells USD after 1 month at 120.10 (as against current Bid rate of 120.25)

FX Market Transactions



DERIVATIVE

- USD/JPY Spot: 120.25/120.50 USD/JPY Forward: 120.10/ 120.35 (1 month)
- JAP IMPORTER WANTS TO HEDGE USD PAYMENT HE WILL MAKE AFTER 1 MONTH
- He approaches a Bank and says I will buy USD after 1 month and I want to book it today
- Bank will buy USD at Spot bid rate: 120.25 JPY
- Bank will sell USD at 1 month Forward offer rate: 120.35 JPY
- Hedging cost: 0.10 JPY. Importer pays this cost and gets USD after 1 month at 120.35 (as against current offer rate of 120.50)
- USD has a higher interest rate. Hence it depreciates. The importer buys USD after 1 month. So he pays 0.15 JPY less for each dollar.

FX Market Transactions



DERIVATIVE

Forward Rate

- Operates on the basis of Theory of Interest Rates Parity
- Anticipated currency exchange rate shifts will be proportional to countries' relative interest rates. OR Expected returns on investment in different currencies should even out.
- Above is true in a floating rate regime and ceteris paribus.
- Japan facing stagnant economic growth. Slow exports also. So it had near zero interest rates (to encourage people to spend), and encouraged a cheap Yen for more exports.
- Yen Carry Trade What is it?
- Yen Carry Trade & When it unwinds
- Japan's \$70 Billion intervention & Rate hike
- Yen's Roller Coaster Ride

FX Market Trxns



DERIVATIVE

FX SWAP TRANSACTIONS

- Do opposite transactions (Buy/Sell or Sell/Buy) simultaneously.
- Combination of a Spot txn done today & a Forward txn done for a future date
- Example: You import crude oil from US and sell them petroleum products after 3 months. Sell INR & Buy USD at Spot Rate today to pay for imports. Contract to sell USD after 3 months at 3 month Forward rate today.
- Hedge against exposure to USD fluctuations.
- Forward Contract: When FX involved only on one leg of the txn
- FX Swap Transaction: When FX involved on both legs
- *Futures & Options are also available as FX derivatives*

Which Interest Rate?



DERIVATIVE

Which Interest Rate to Use for FX Forward & Swap Transactions

- A Bit of History - LIBOR: London Interbank Offer Rate.
- Benchmark interest rate at which major banks claimed they could borrow unsecured funds from other banks in the London Interbank Market.
- *Each day a panel of major banks was asked: "At what rate could your bank borrow funds today?" Banks submitted estimates. The benchmark was then calculated from these submissions and announced. **NO INDEPENDENT VERIFICATION OF DATA.***
- Here's What Went Wrong With LIBOR
- How Barclay's Manipulated LIBOR UBS & Royal Bank of Scotland also involved.
- LIBOR Scandal Three Things to Know

Which Interest Rate?



DERIVATIVE

Which Interest Rate to Use for FX Forward & Swap Transactions

- Different Benchmarks for different currencies based on actual market transactions data.
- USD: SOFR Secured Overnight Financing Rate
- GBP: SONIA Sterling Overnight Index Average
- EUR: € STR Euro Short Term Rate
- JPY: TONAR Tokyo Overnight Average Rate
- INR: MIBOR Mumbai Interbank Offered Rate

Foreign Exchange Flows



Changes in FX Flows

DUE TO CHANGES IN FLOW IN CAPITAL ACCOUNT

- Both way movement: Inflow & Outflow
- Types
 - FDI Foreign Direct Investment: To set up new business or invest in existing business.
 - FPI Foreign Portfolio Investment: To invest in a country's stock and bond markets. By Financial Institutions & NRIs
 - Bank Investments: Investment in NRO & NRE bank deposits
 - External Commercial Borrowings (ECB): Bonds sold abroad
 - Purchase/sales of assets like Land/ IP rights etc.
 - Foreign Aid or Grant for capital expenditure Ex. building Delhi Metro

RBI confirms record \$94.5b gross FDI in FY26, net inflows just \$7.7b

Reserve Bank of India data shows gross FDI inflows surged 17% year-on-year to \$94.5 billion in FY26, the highest ever for India. However, record repatriations of \$53.6 billion and outward investments by Indian companies worth \$33.3 billion reduced net inflows to \$7.7 billion, about 8% of the gross. This is still a marked improvement from the sub-\$1 billion net inflows in FY25, breaking a four-year decline in net FDI. CNBC-TV18 + 2



THE  HINDU

FPI exodus continues, ₹62,800 crore pulled out from equities in first fortnight of June

According to NSDL data, FPIs have remained net sellers in every month of 2026 except February

RBI swap windows could attract \$40-50 billion forex inflows in FY27

According to a latest note by Motilal Oswal, the RBI's newly announced FCNR(B) and ECB swap facilities could attract \$40-50 billion of foreign currency inflows into the banking system during FY27. The brokerage said the measures are expected to support deposit mobilisation, strengthen foreign exchange reserves and ease liquidity pressures, while creating opportunities for banks with strong overseas franchises and NRI customer bases.



ETBFSI Staff • ETBFSI

Updated On Jun 12, 2026 at 04:39 PM IST





Changes in FX flows

DUE TO CHANGES IN FLOWS IN CURRENT ACCOUNT

- Both way movement: Inflow & Outflow
- Due to trade (export/ import) of goods & services
- Due to Income (Compensation of employees, Investment income)
- Due to Current Transfers: Remittances, Foreign Aid & Grants for consumption (Feeding people affected by floods)
 - Inward Remittances: Sent to home country by citizen's working abroad for their families and relatives



A longer Gulf conflict could squeeze remittances to India

K Johny & Dilasha Seth, ET Bureau • Last Updated: Mar 21, 2026, 01:59:53 PM IST

Preferred on G



More than a third of India's remittances, worth Rs 3.74 lakh crore (\$40 billion) in 2023-24, came from Gulf nations such as the UAE, Saudi Arabia, Qatar, Kuwait and Oman, according to the latest RBI data. The UAE, which counts Indians as the single-largest expatriate group, was the second-largest source of remittances in FY24 at 19.2%, while Gulf nations together accounted for 38%.

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Remittances beat FDI: India's biggest export isn't software or steel—it's its people

India received a record \$135.4 billion in remittances in FY25—far above FDI inflows. But rising instability in West Asia threatens this lifeline, pushing India to diversify global job markets.

LARGEST REMITTANCE
RECEIVING COUNTRY IN
THE WORLD



Anoop Verma • ETGovernment

Published On Apr 27, 2026 at 06:49 AM IST





Changes in FX flows

DUE TO POLITICAL UPHEAVALS/ STRANGE POLICIES

- <https://www.xe.com/currencycharts/?from=USD&to=BDT&view=10Y>
- USD/ Bangladeshi Taka 10 Yr chart: Regime Change + Eco issues
- USD/ Ukrainian Hryvnia 10 Yr chart: War
- USD/ Turkish Lira 10 Yr Chart: Lowered interest rates when inflation was rising

Changes in FX flows



CHANGES IN CENTRAL BANK FX RESERVES. Due to Central Bank Intervention

- <https://data.rbi.org.in/DBIE/#/dbie/home>

Item	As on Jun. 05, 2026		Variation over					
			Week		End-March 2026		Year	
	₹ Cr.	US\$ Mn.	₹ Cr.	US\$ Mn.	₹ Cr.	US\$ Mn.	₹ Cr.	US\$ Mn.
	1	2	3	4	5	6	7	8
1 Total Reserves	6470564	681610	-13114	-711	-83297	-9498	502461	-15046
1.1 Foreign Currency Assets #	5158761	543444	-30803	-2704	-78606	-8839	124219	-44243
1.2 Gold	1087630	114575	17686	1975	-6680	-820	351851	28687
1.3 SDRs	178134	18765	-3	18	1545	144	18181	94
1.4 Reserve Position in the IMF	46039	4826	7	0	444	18	8211	416

- Foreign Currency Assets: Assets held in major foreign currencies by RBI (USD, GBP, JPY etc.)
- SDR Special Drawing Rights
- Reserve Position in IMF: Like SDR without payment of interest



Balance of Payment

- Statistical statement of all economic transactions between Indian residents with the rest of the world.
- Transactions in Capital & Current Account added to see if amount of FX is in excess or deficit.
- Balanced by Central Bank Reserves.
- Similar to accounting in Balance Sheet of a company
- India BoP, May 22, 2026: https://www.rbi.org.in/Scripts/BS_ViewBulletin.aspx?Id=24221
- Row 1, Point 1, Point 2, Point 3, Point 4 & Point 4.2

FX Demand & Supply, Equilibrium Exch Rate & Change in FX rates



A Sampler Only

- Topic Requires In Depth Study. Is complex.
- FX rates are impacted by multiple factors.
 - Changes in supply & demand in goods & services market
 - Money market (supply demand of money, bonds etc.)
 - Factors like interest rates, inflation, rate of economic growth, political stability etc.
 - Government & Central Bank policies
 - Ease of capital flows (in and out) for investments in assets
 - Speculation/ expectation about future: Inflation, National Debt etc
- We will just dip our toes in the pond and get a sense of how the water feels.

Types of Exch Rate Regimes



- Fixed Exchange Rate
 - Exch Rate pegged to another currency. Saudi/ UAE. Peg is USD.
- Free Floating/ Floating Exchange Rate
 - Allow the FX rate to change depending on different factors. US, Euro, Japan, UK, Sweden etc
- Managed Exchange Rate
 - Aggressive intervention in the FX markets to keep it around a desired value. This value can change. China.
- India ?
- <https://worldpopulationreview.com/country-rankings/exchange-rate-regime-by-country>



Exch Rate Regime

- How can the Govt/Central Bank influence FX rates?
 - **Fiscal Policy:** Increase or decrease expenses to impact demand of goods in the economy. Budget of the government & other policies like Tariffs.
 - **Monetary Policy:** By changing interest rates or supply of money in the economy. By Central Bank. Often on directions of the Govt.
 - **Direct intervention** by Central Bank and Govt owned Banks. Buy and sell domestic/ foreign currency
 - **Changes in policies** leading to FX flows for investments etc. Controls on flow of capital in and out of the country

Types of Exch Rate Regimes



Readings for those interested

<https://www.brookings.edu/articles/the-renminbi-the-political-economy-of-a-currency/>

<https://www.brookings.edu/articles/chinas-currency-policy-explained/>

<https://www.elibrary.imf.org/view/journals/024/1979/004/article-A002-en.xml>

https://www.congress.gov/crs_external_products/RL/PDF/RL31204/RL31204.5.pdf

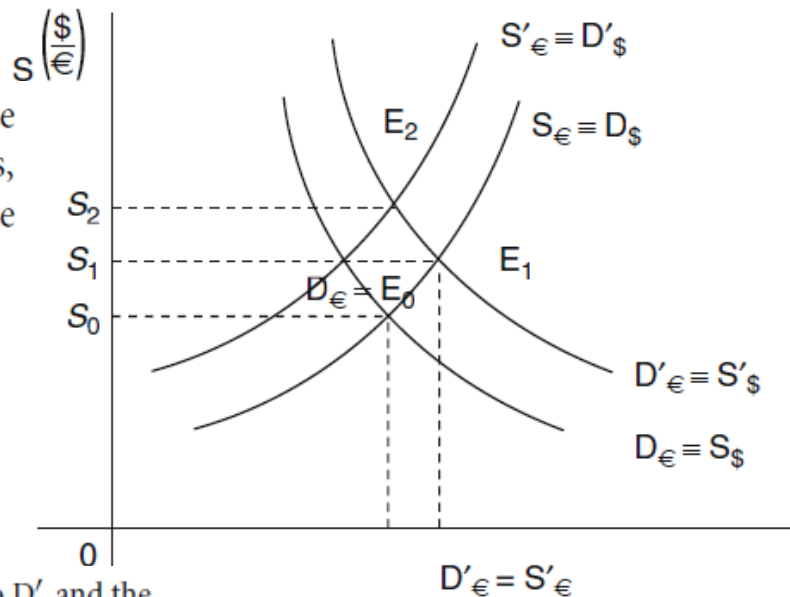


Exch Rate Changes: BoP Approach

$$B \text{ of } P_t = CA \left(\frac{S_t P_t^*}{P_t}, Y_t, Y_t^*, \Pi_t, \Pi_t^* \right) + KA(i_t - i_t^*)$$

where, CA =current account, S_t =spot exchange rate, P_t =domestic price level, P_t^* =foreign price level, Y_t =real income, Π_t =shift factor (tariffs, subsidies, interventions, etc.), KA =capital account, $i_t - i_t^*$ =interest rate differential, and an asterisk (*) denotes the foreign variables.

Assuming Floating Exchange Rates



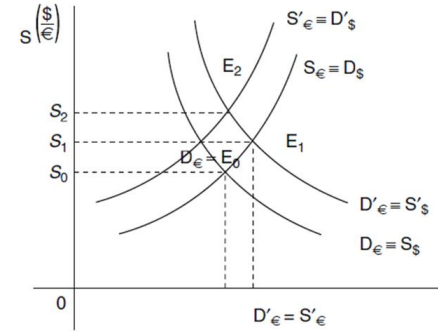
Note: An increase in the demand for euros (€) shifts the demand schedule from D_e to D'_e and the original equilibrium exchange rate s_0 (at the intersection of the demand and supply schedules) increases to s_1 . The euro is appreciated and the US dollar is depreciated. A reduction in supply of euros is—a shift of the supply schedule to the left from S_e to S'_e (from E_1 to E_2).



Exch Rate Changes: BoP Approach

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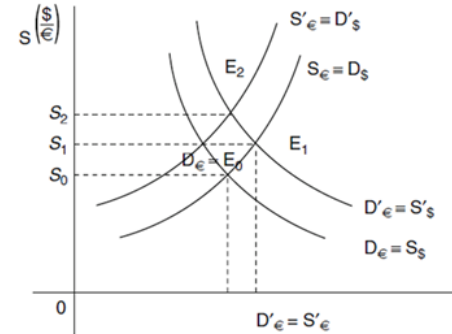
- Increase in domestic interest rates: FX will flow in to invest in domestic currency assets. Increase in Demand for Domestic Currency which will appreciate. **ceteris paribus**
- Increase in foreign interest rates: Opposite of above. Domestic currency will depreciate. **ceteris paribus**



Exch Rate Changes: BoP Approach

$$B \text{ of } P_t = CA \left(\frac{S_t P_t^*}{P_t}, Y_t, Y_t^*, \Pi_t, \Pi_t^* \right) + KA(i_t - i_t^*)$$

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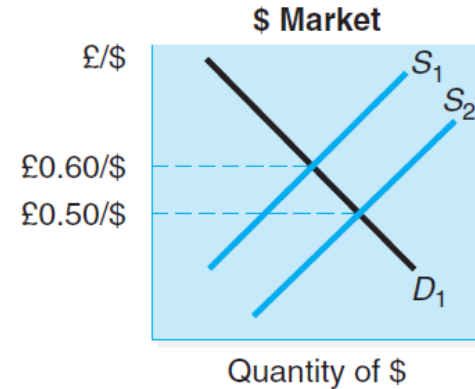
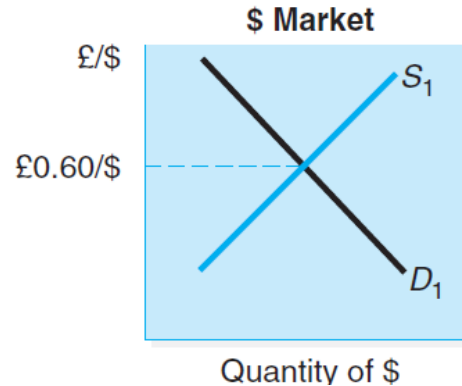
- Increase in Eco Growth or Income: Demand for imported goods will increase. Demand for FX will increase. Domestic currency will depreciate. **ceteris paribus**
- Increase in Tariffs on imports: Imports will reduce. Exports remain same as earlier. FX flow (Export – Import) is higher. Domestic currency will appreciate. USA? **ceteris paribus**
- Increase in Productivity: Exports become cheaper. Hence increase in exports. More FX flow. Domestic currency will appreciate. CHINA? **ceteris paribus.**
- Government could also give subsidies or other help to increase exports.



Changes in Exch Rate

- US imports more from UK then it exports to them. Hence has a Current Account Deficit with respect to UK
- US Govt wants to depreciate USD so that its exports to UK become cheaper and imports become expensive. And deficit reduces.
- One solution: US Fed sells USD. (and buys GBP)

Central Bank Intervention



To make the dollar depreciate, the Fed should sell dollars, resulting in a lower equilibrium exchange rate.



A FEW NOTES

- LONG TERM FX RATES: Theory of Purchasing Power Parity
- REAL EXCHANGE RATE: Over or undervalued currency
- SHORT TERM FX RATES:
 - BoP Approach
 - Asset Market Approach: Investors choose between domestic & foreign assets based on expected returns. FX rate changes to equalize expected returns
 - Interest Rate Parity: FX rates change based on difference in interest rates. Refer example solved earlier on Forward Rate.



A FEW NOTES

- SHORT TERM FX RATES:
 - Monetary Policy Approach: Treats FX changes as affected by money supply, money demand, income, inflation. If a country's money supply grows much faster than output, prices rise and the currency depreciates.
 - Inflation & Interest Rates: Impact FX as well as Bonds
- We used these approaches at various points in our class discussions.
- Each of these approaches make certain assumptions and are useful to explain FX movements in different situations.



Thank You!